

	R – contact force on ground due to box S – gravitational force on Earth due to box
5	B Conservation of linear momentum : Total initial momentum = mv Total final momentum of Option A $= mv - mv = 0$ By velocity of approach and separation, KE not conserved for option A Total final momentum of Option B

$$= m(v + 0) = mv$$

KE is conserved for option B.

Total final momentum of Option C

$$= m\left(\frac{3v}{4} - \frac{v}{4}\right) = \frac{1}{2}mv$$

By velocity of approach and separation, KE is conserved for option C

Total final momentum of Option D

$$= m\left(\frac{v}{2} + \frac{v}{2}\right) = mv$$

By velocity of approach and separation, KE not conserved for option D

6

D

Y